25AIMLPCC301ESE

13 OCT 2026

Seat No. : [redacted]

Student Name : [redacted]

St. John College of Engineering and Management, Palghar, Maharashtra
Autonomous Institute

End Semester Examinations, October 2026

B.Tech. Artificial Intelligence and Machine Learning -

SEMESTER III

25AIMLPCC301 - Linear Algebra and Optimization for AI

Time : 2.00 Hours **Maximum Marks : 60**

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

		Attempt all Questions	Mrk	Course Outcome	RBT Level
1.	a	i. Using Cauchy Schwartz inequality prove that $\frac{b+c}{a}+\frac{c+a}{b}+\frac{a+b}{c}\geq 6$.	5	CO1	L3
		ii. Find all vectors in R^3 of Euclidean norm 1 that are orthogonal to vectors $u_1=(1,1,-1), u_2=(1,1,0)$.	5	CO1	L3
		OR			
	b	Find orthonormal basis for subspaces of R^3 of $S=\{(1,0,0),(3,7,-2),(0,4,1)\}$ by applying Gram Schmidt process.	10	CO1	L4
2.	a	i. Find the matrix represented by $A^9-6A^8+10A^7-3A^6+A+I$ for $A=\begin{pmatrix}1&2&3\\-1&3&1\\1&0&2\end{pmatrix}$	5	CO2	L4
		ii. Verify Cayley Hamilton theorem for $\begin{pmatrix}2&-1&1\\-1&2&-1\\1&-1&2\end{pmatrix}$ and find A^{-1}..	5	CO2	L4
		OR			
	b	Find the eigen value and eigen vectors of $\begin{pmatrix}3&-1&1\\-1&5&-1\\1&-1&3\end{pmatrix}$.	10	CO2	L4
3.	a	i. Reduce $2x^2-2y^2+2z^2-2xy-8xz+6zx$ to canonical form under congruent transformations.	5	CO3	L4
		ii. Determine the nature of $6x_1^2+49x_2^2+51x_3^2-4x_1x_2-82x_2x_3+20x_1x_3$.	5	CO3	L3
		OR			
	b	Determine whether the matrix $A=\begin{pmatrix}2&1&1\\1&2&1\\0&0&1\end{pmatrix}$ is diagonalizable.If yes, then find the diagonal and modal matrix.	10	CO3	L4
4.	a	i. Compute the standard form of the following LPP. 1) $Maximize\ z=3x_1+2x_2+5x_3$ $subject\ to, 2x_1-3x_3\leq 3$ $x_1+2x_2+3x_3\geq 5$ $3x_1+2x_3\geq 2$ $x_1,x_2,x_3\geq 0$	5	CO4	L3

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		2) $Maximize\ z = 2x_1 + 3x_2 + 5x_3$ $subject\ to, 3x_1 - 2x_2 \leq 6$ $2x_1 + 3x_2 + 5x_3 \geq 12$ $3x_1 + 4x_3 \leq 5$ $x_1, x_2 \geq 0,\ x_3$ *is unristricted*			
		ii. Analyze the given LPP and find all basic solutions of $Maximize\ z = x_1 + 3x_2 + 3x_3$ $subject\ to, x_1 + 2x_2 + 3x_3 = 4$ $2x_1 + 3x_2 + 5x_3 = 7$	5	CO4	L4
		OR			
	b	Apply Simplex method to solve $Maximize\ z = 5x_1 + 4x_2$ $subject\ to, 6x_1 + 4x_2 \leq 24$ $x_1 + 2x_2 \leq 6$ $-x_1 + x_2 \leq 1$ $x_2 \leq 2$ $x_1, x_2 \geq 0$	10	CO4	L3
5.	a	i. Analyze the given function to find relative maxima or minima of $Z = x_1^2 + x_2^2 + x_3^2 - 4x_1 - 8x_2 - 12x_3 + 100.$	5	CO5	L4
		ii. Apply dual simplex method to solve $Minimize\ z = 2x_1 + 2x_2 + 4x_3$ $subject\ to,$ $2x_1 + 3x_2 + 5x_3 \geq 2$ $3x_1 + x_2 + 7x_3 \leq 3$ $x_1 + 4x_2 + 6x_3 \leq 5$ $x_1, x_2, x_3 \geq 0$	5	CO5	L3
		OR			
	b	Analyze the optimization problem using the method of Lagrange's multiplier to solve $Optimize\ z = 10x_1 + 4x_2 + 4x_1x_2 - x_1^2 - 5x_2^2$ $subject\ to,$ $x_1 + x_2 = 6$ $x_1, x_2 \geq 0$	10	CO5	L4
6.	a	i. Apply KT method to solve $Maximize\ z = 10x_1 + 4x_2 - 2x_1^2 - 2x_2^2$ $subject\ to,$ $2x_1 + x_2 \leq 5$ $x_1, x_2 \geq 0$	5	CO6	L3
		ii. solve $Maximize\ z = 8x_1 + 10x_2 - x_1^2 - x_2^2$ by Kuhn Tucker method. $subject\ to,$ $3x_1 + 2x_2 \leq 6$ $x_1, x_2 \geq 0$	5	CO6	L3
		OR			
	b	Analyze the optimization problem using the method of Lagrange's multiplier to solve $Optimize\ z = 8x_1 + 9x_2 - x_1^2 - x_2^2$ $subject\ to,$ $2x_1 + 3x_2 = 12$ $5x_1 + 2x_2 = 19$ $x_1, x_2 \geq 0$	10	CO6	L4

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* All the Best *